

Subject Description Form

Subject Code	LSGI3262
Subject Title	Laser Scanning
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are: • To provide students with an overview of the theoretical and practical aspects of laser scanning • To provide students with an understanding of the applications of the technology • To enable students to become skillful in operating laser scanning equipment, and apply the techniques to practical problems • Student's critical thinking, team spirit, problem-solving, leadership, and presentation skills will be developed through group project
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Explain the basic principles and operations of laser scanning systems b. Understand and describe the applications of laser scanning c. Compare and identify appropriate laser scanning systems for real-world applications d. Perform laser scanning operations, including mission planning, data acquisition, and data processing, for application cases e. Control and evaluate the quality and accuracy of laser scanning products
Subject Synopsis/ Indicative Syllabus	A. Introduction to laser scanning - Basic math and physics concepts used in laser scanning (e.g. vector, solid angle, photons, phase and amplitude, radiant power, intensity, apparent reflectance) - Electromagnetic wave propagation and laser pulse generation - Basic principles of laser scanning - Laser ranging methods and measurements B. Instrumentation and laser scanning systems - Mechanism and working principles of laser scanner and other components, including inertial measuring unit, inertial navigation system, and global positioning system - Types of lidar sensors: discrete return, waveform, single-photon detection, and solid-state lidar.

	- Spaceborne, airborne, terrestrial, mobile sensor platforms, and handheld laser scanning systems C. Data acquisition, processing, and management - Coordinate system and datum - Geometry of laser scanning system - Project planning, design, and operation, such as choosing sensor platforms for various purposes - Laser interaction with various target types. - Post data processing, including trajectory processing, data integration, georeferencing, noise removal, classification, and point matching - Data storage and management D. Accuracy and quality assurance - Error sources in laser scanning - Methods and standards for quality control - System calibration - Accuracy evaluation E. Applications of laser scanning - Digital Surface Model (DSM), Digital Terrain Model (DTM), and Digital Height Model (DHM) generation - 3D Model Generation, including triangularization, and voxelizations - Feature extraction and matching - 3D modelling, engineering applications, high-definition mapping, indoor applications, forestry, and archaeological excavation						
Teaching/Learning Methodology	Teaching and learning materials will be uploaded online for students to download easily. Contact hours will be used for formal lectures, practical sessions, and a group project. Lectures will be delivered to introduce the subject materials. Lab sessions and individual assignments will be used to reinforce the theories, methodology and practical skills introduced during the lectures, so as to enable students to gain a deeper understanding of the principles and techniques. A group project will be used to enhance critical thinking, team spirit, problem-solving, leadership, and presentation skills.						
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e
	1. Practical assignments and group project	50			✓	✓	✓

	2. Phase test	15	✓	✓	✓		
	3. Final test	35	✓	✓	✓		✓
	Total	100%					
	Students' performance will be assessed based on practical assignments, group project, phase test and final test. Practical assignments are used to assess students' practical skill in processing laser scanning data. A group project is used to assess students' ability and practical skills in planning activity, fieldwork operation of laser scanning equipment, data processing, and report preparation for surveying tasks. The written phase test and final test will test the students' independent skills of expression, knowledge and understanding of the theories, techniques, and methods of laser scanning.						
Student Study Effort Expected	Class contact:						
	▪ Lectures					26 Hrs.	
	▪ Practical sessions					26 Hrs.	
	Other student study efforts:						
	▪ Practical assignments					24 Hrs.	
	▪ Group project					32 Hrs.	
	Total student study effort					108 Hrs.	
Reading List and References	1. Vosselman, G., & Maas, H. G. (2010). <i>Airborne and Terrestrial Laser Scanning</i>. Dunbeath: Whittles Publishing. 2. Shan, J., & Toth, C. K. (Eds.). (2018). <i>Topographic Laser Ranging and Scanning : Principles and Processing</i> (Second edition.). Boca Raton, FL: CRC Press. 3. Heritage, G. L., & Large, A. R. (2009). <i>Laser Scanning for the Environmental Sciences</i>. Wiley. 4. Baltsavias, E. P. (1999). Airborne Laser Scanning: Basic Relations and Formulas. <i>ISPRS Journal of Photogrammetry and Remote Sensing</i>, 54(2-3), 199-214. 5. Land and Engineering Survey Board, The Government of the Hong Kong Special Administrative Region. (2023). <i>Engineering Survey Practice Guides for Works Projects</i> (Second edition.). 6. The International Organization for Standardization (2018). <i>Optics and optical instruments — Field procedures for testing geodetic and surveying instruments — Part 9: Terrestrial laser scanners</i>. 7. Department of Land Surveying and Geo-Informatics (LSGI) (2020). <i>Specifications for Nondestructive Testing, Surveying, Imaging and Diagnosis for Underground Utilities 1,3 Laser Scanning Survey</i>. https://www.polyu.edu.hk/lsgi/uusspec/en/Publications/						